Introduction to cloud computing

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The power of the cloud

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Cloud service models

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Cloud deployment models

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Cloud deployment models

- Important decision in cloud adoption
- How much control do you need over your cloud environment?
- Three main types: **private, public, and hybrid**

Private cloud

Cloud infrastructure is designated for exclusive use by its tenants.

Private clouds are accessed by a network link.

Pros: Direct control of resources and data

Cons: More upfront investment

Unlike on-premise, private cloud uses virtualization for on-demand compute resources and can be off-premises.

Public cloud

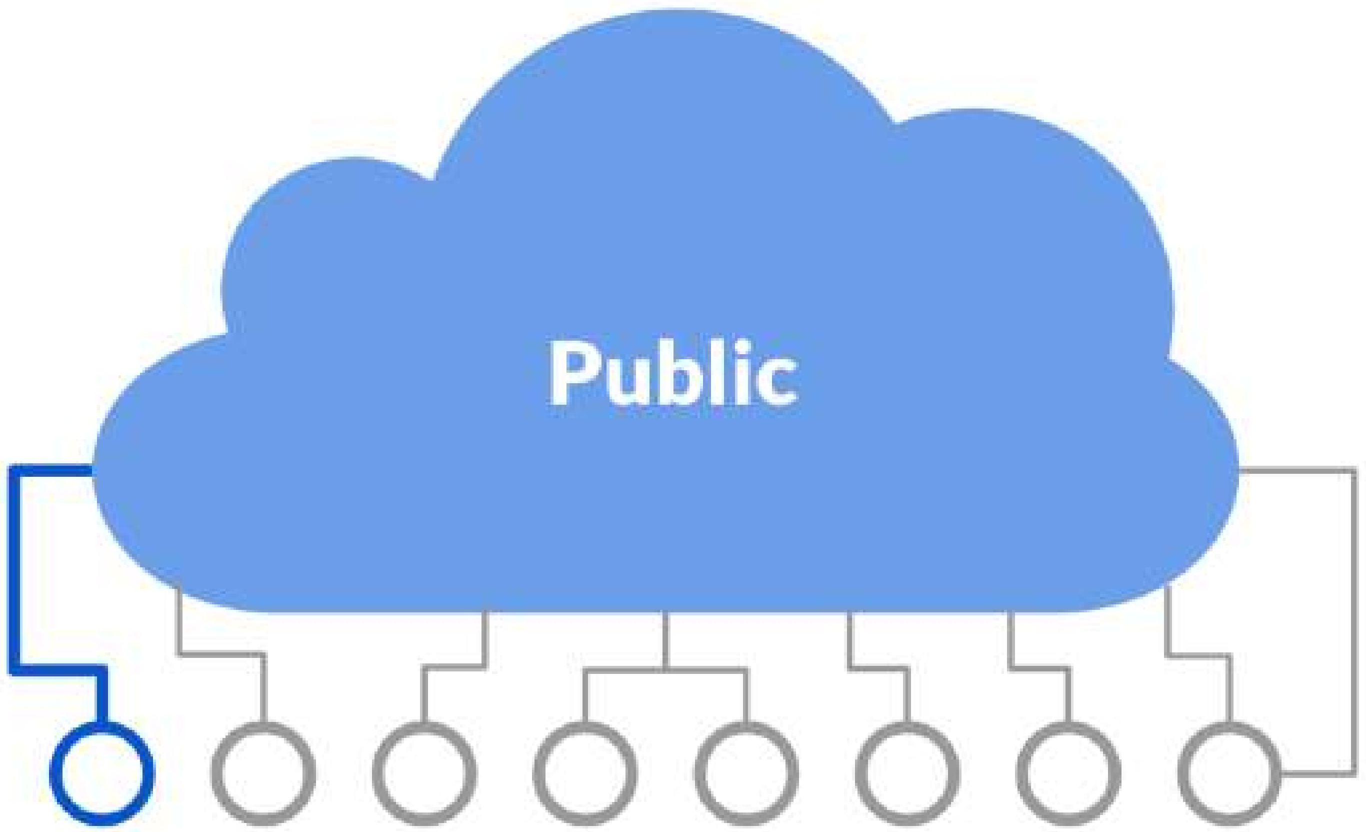
Cloud infrastructure is shared and open for use by the general public. It's owned and managed by a cloud service provider.

Public clouds are Internet accessible.

Pros:

- Get started quickly with minimal investment
- Easier to scale

Cons: No access to data center and hardware

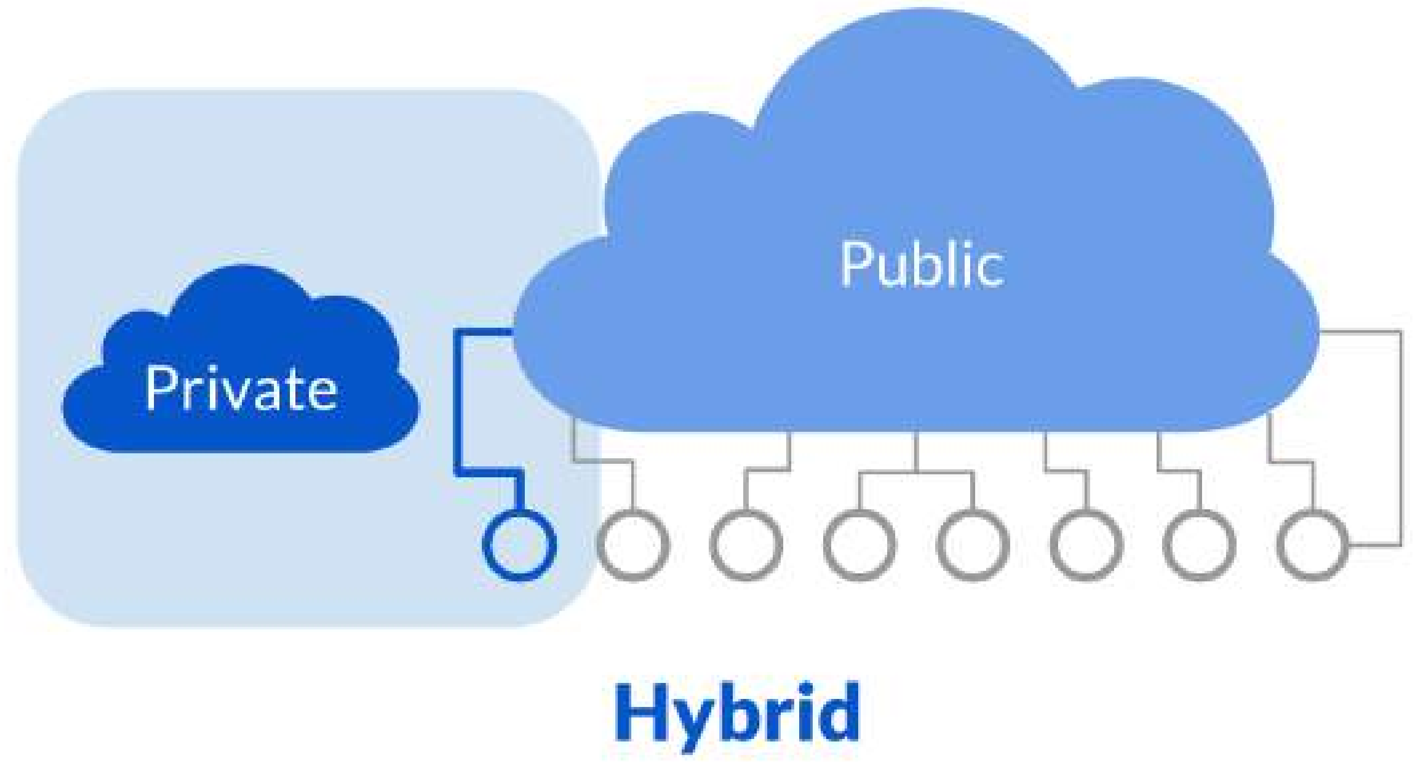


Hybrid cloud

Organization uses a combination of two or more distinct models.

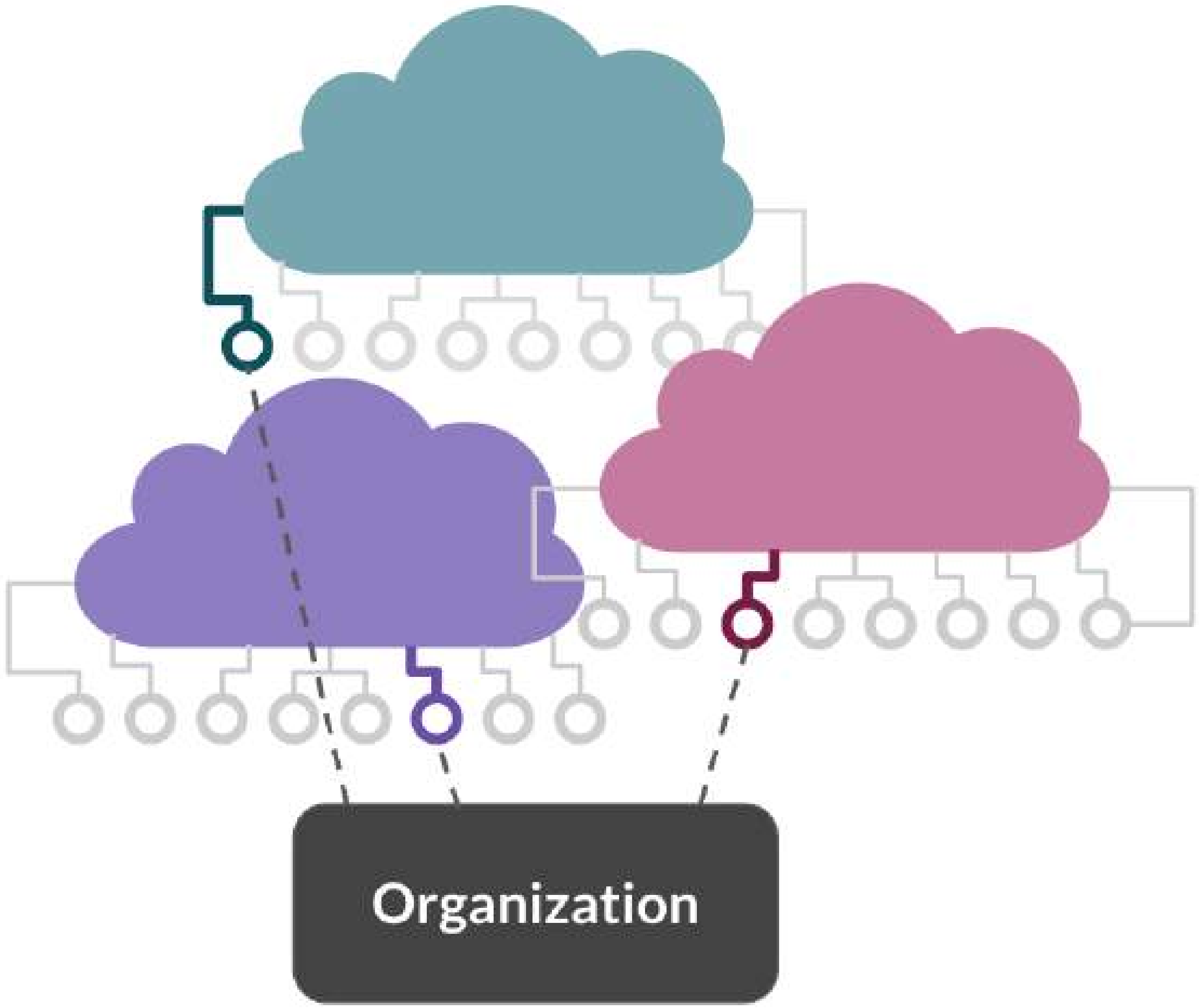
Use cases:

- Store sensitive data on the private cloud and use application on public cloud for analytics
- **Cloud bursting:** when private cloud hits capacity, temporarily move overflow to the public cloud to avoid disruption of service



Other deployment models

- **Multicloud**
  - Combination of different cloud provider services
  - Flexibility on pricing plans and service offerings
  - No reliance on one vendor



Other deployment models

- **Community**
  - Infrastructure shared by a specific community for exclusive use
  - Common interest or concern, e.g., security, jurisdiction, mission
  - Can be managed and hosted internally or externally

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Regulations on the cloud

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General Data Protection Regulation (GDPR)

- Regulates how personal data is collected, processed, and stored from users in the EU
- Examples:
  - Users must explicitly consent to data collection
  - Notify users of any data breaches
  - Personal data information must be encrypted, anonymized, and/or pseudonymized
  - **Personal data can't leave EU borders, unless you can guarantee the same level of protection**
- Fine: 20 million Euros or up to 4% of the worldwide annual revenue

What is personal data?

> Personal data is any information that relates to an identified or identifiable living individual. Different pieces of information, which collected together can lead to the identification of a particular person, also constitute personal data. [1]

- Includes: *home address, first name, last name, email address, location data, IP address, racial or ethnic origin, political opinions, sexual orientation, health related data*
- Personally identifiable information (PII)

[1] https://ec.europa.eu/info/law/law-topic/data-protection/reform/what-personal-data_en

Other regulations

- Brazil's Lei Geral de Proteçao de Dados (LGPD)
- California's Consumer Privacy Act (CCPA)
- USA's Health Insurance Portability and Accountability Act (HIPAA)
- Japan's Act on Protection of Personal Information
- Thailand Personal Data Protection Act (PDPA)
- Canada's Personal Information Protection and Electronic Documents Act (PIPEDA)

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Cloud roles

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Most in-demand skill: cloud computing

Familiar data roles and the cloud

- **Data scientist**
  - Run computationally expensive analyses on the cloud
- **Machine learning scientist**
  - Train and deploy machine learning models on the cloud
- **Data engineer**
  - Build pipelines on the cloud to ingest, transform, and store big data
- **Data analyst**
  - Access data on the cloud via business intelligence tools

Creation of new cloud roles

- Cloud architect
- Cloud engineer
- DevOps engineer
- Security engineer

Cloud architect

- Solutions architect for the cloud
- Design cloud infrastructure for a given business problem
- Plan the deployment of the infrastructure
- Ensure scalability and optimized for cost

Cloud engineer

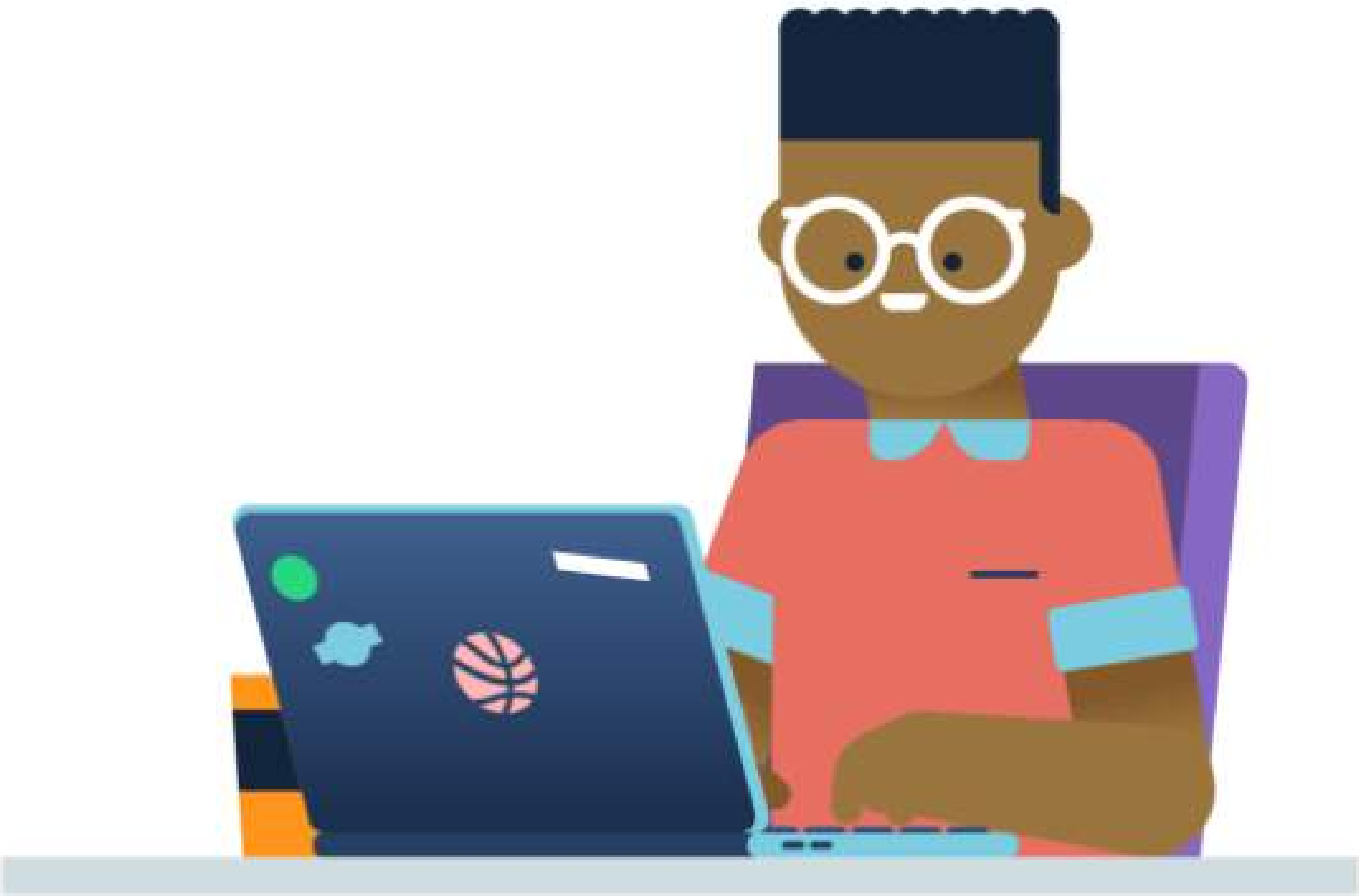
- Build, maintain and monitor cloud services
- Migrating operations to the cloud

DevOps engineer

- Software **Dev**elopment + IT **Op**erations
- Infrastructure as code
- Ensure the reliability, availability, and scalability of the cloud through software development and automation

Security engineer

- Spec security requirements
- Test and assess security of data on the cloud
- Responsible for protecting organization and user data



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An overview of providers

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Overview

How this chapter should be understood:

- overview of **main cloud providers**
- overview of their **market position**
- overview of their respective **key services**
- overview of their **strengths**
- examples of **customers**
- **case study**

Overview

How this chapter should be understood:

- overview of **main cloud providers**
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- examples of **customers**
- **case study**

The players

Microsoft Azure

ORACLE CLOUD

salesforce

Google Cloud

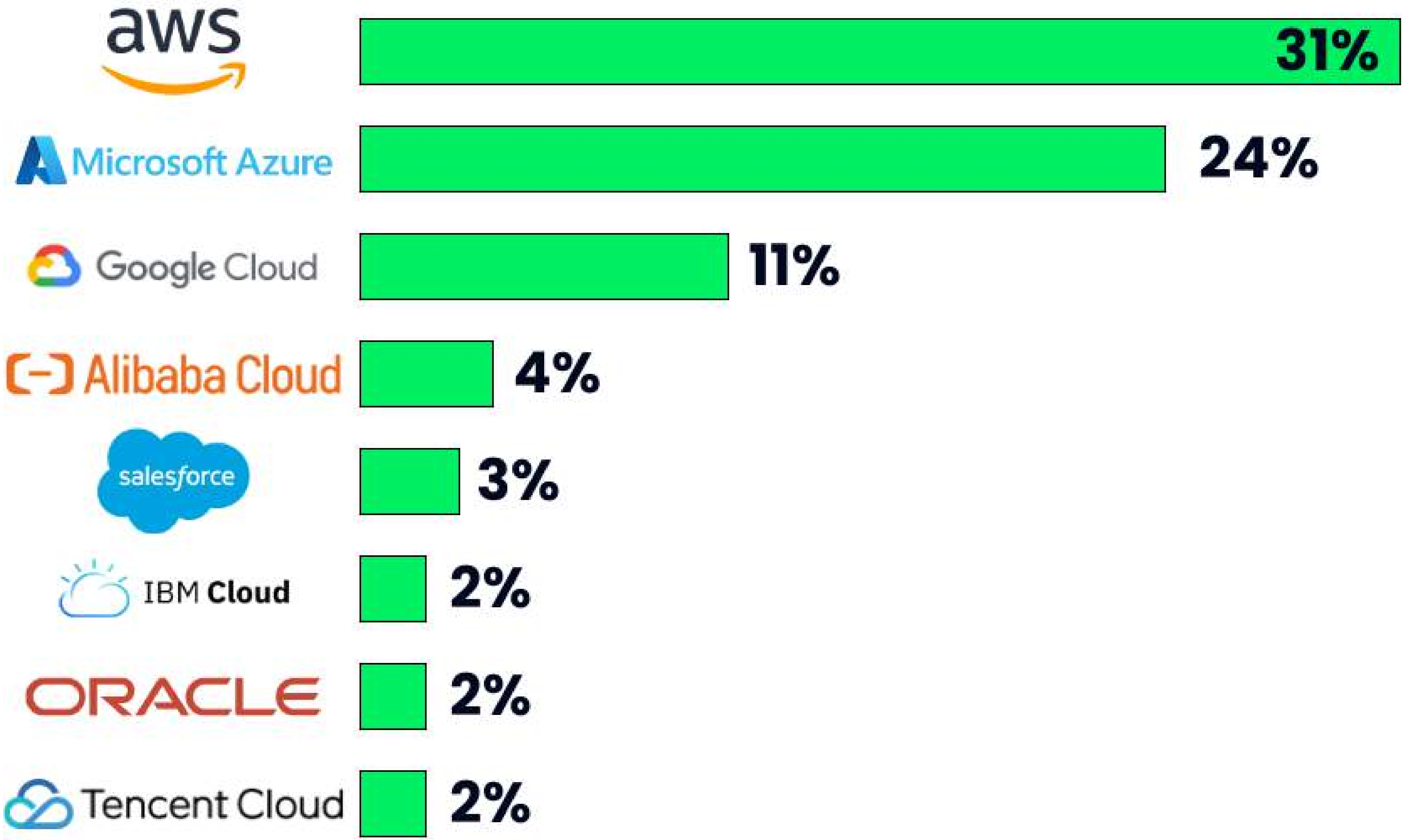
aws

Alibaba Cloud

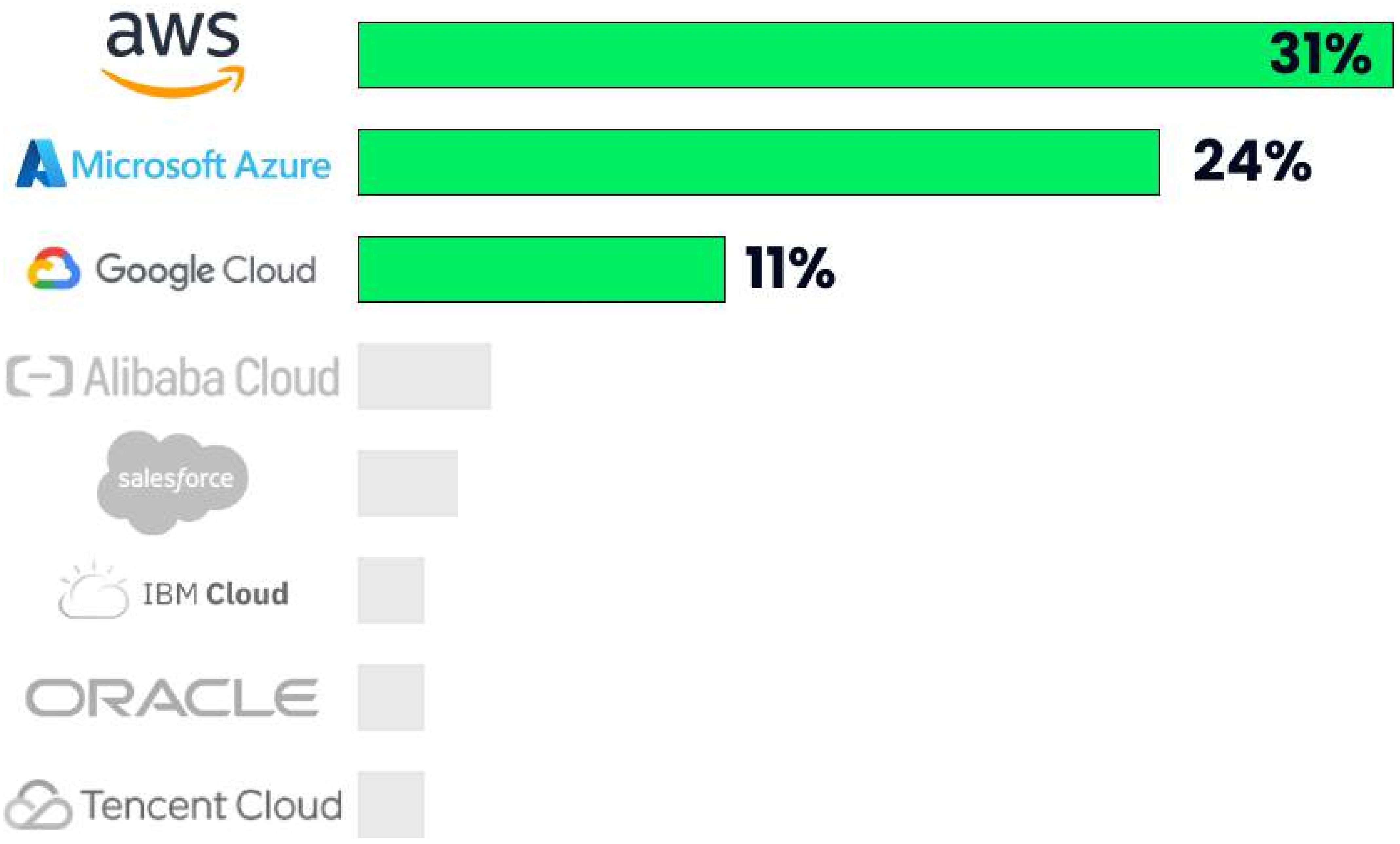
Tencent Cloud

IBM Cloud

Market share



Market share



The rise of cloud computing

- Cloud computing services vital for modern companies
- IaaS and PaaS offer significant benefits
- Enable agility, efficiency, innovation
- Reduce costs, focus on core business

Making a choice

- Best cloud provider meets company needs
- Leverage cloud specialists' knowledge
- Contact providers directly

Making a choice

- Consider current infrastructure and data center costs
- Evaluate costs for managing hardware and storage
- Assess costs for app depreciation, migration, or rebuild for cloud
- Consider hiring cloud specialists, benefits to company and customers, and potential cloud migration risks

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Amazon Web Services

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AWS and the market

- AWS launched in **2006** (Google Cloud in **2008**, Microsoft Azure in **2010**)
- Breadth of services:
  - Computing
  - Storage
  - Analytics
  - Security and enterprise applications
  - Machine learning
- Market share: **31%**

AWS professional cloud services

AWS Simple Storage Service (S3)

AWS professional cloud services

AWS Simple Storage Service (S3)

AWS Elastic Compute Cloud (EC2)

AWS professional cloud services

AWS Simple Storage Service (S3)

AWS Elastic Compute Cloud (EC2)

AWS Relational Database Service (RDS)

AWS professional data services

- **Redshift** (analytics - data warehousing)

AWS Redshift

AWS professional data services

- **Redshift** (analytics - data warehousing)
- **Kinesis** (real time data movement and analytics)

AWS Redshift

AWS Kinesis

AWS professional data services

- **Redshift** (analytics - data warehousing)
- **Kinesis** (real time data movement and analytics)
- **SageMaker** (predictive analytics and machine learning)

AWS Redshift

AWS Kinesis

AWS SageMaker

AWS customers

AWS case study

Company: NerdWallet

Problem: Takes too long to deploy machine learning models

Solution:

- Amazon Sagemaker (cloud machine learning platform gathering machine learning processes)

AWS case study

Improvements:

- Reduce training times to days
- Reduce training costs by 75%
- Modernized data science engineering practices

[1] https://aws.amazon.com/solutions/case-studies/

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Microsoft Azure

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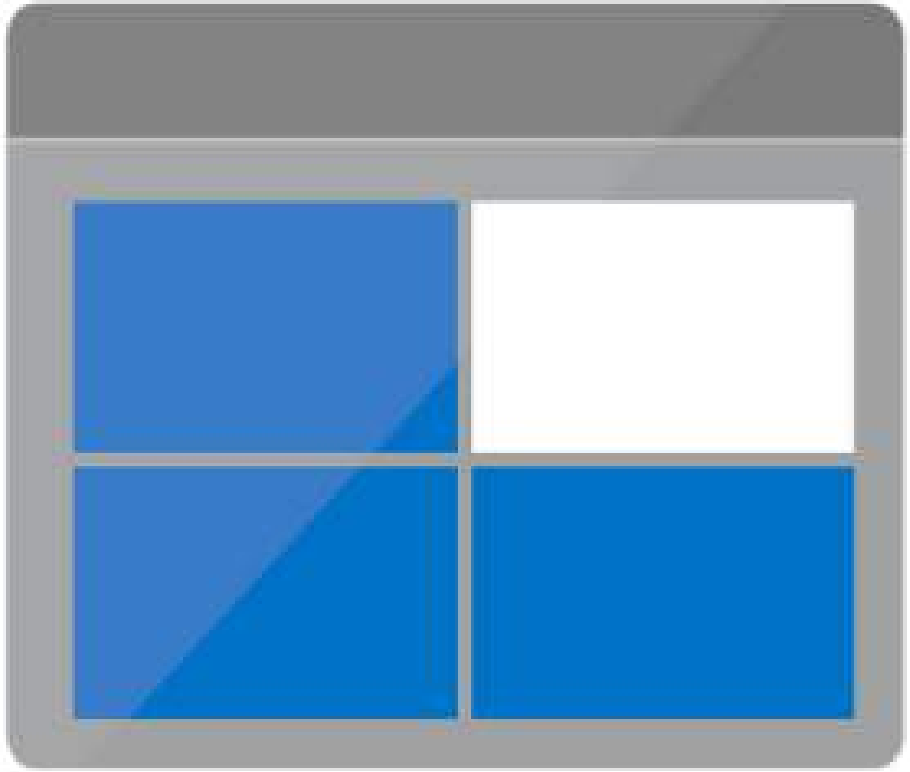
Azure and the market

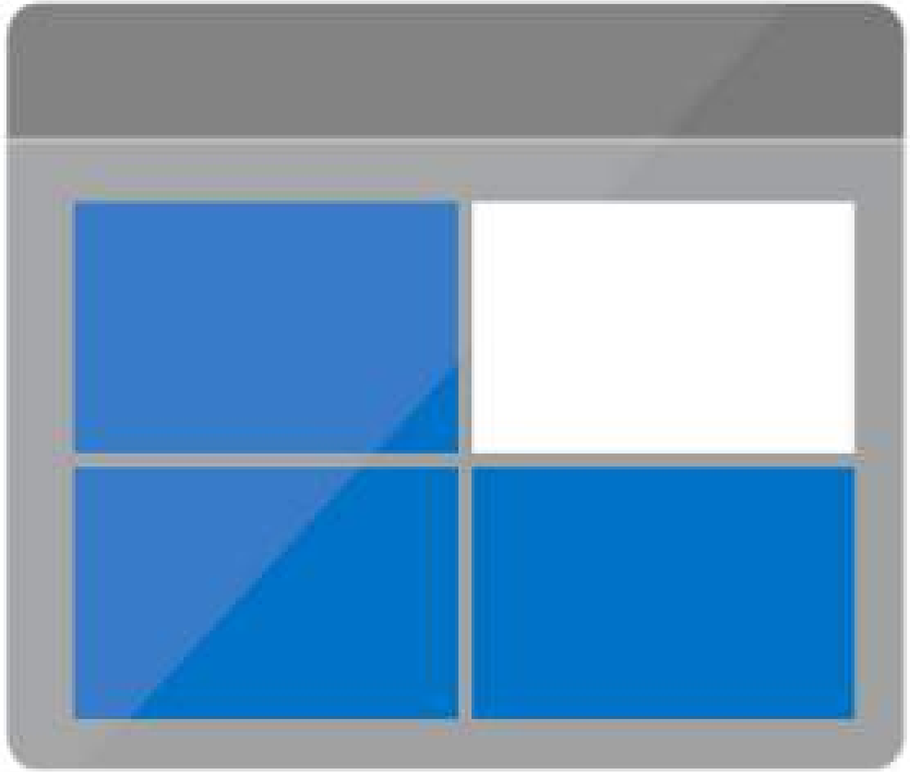
- Integration with Microsoft products
- Benefits from customer loyalty, top-of-mind choice
- Market share: **24%**

Azure cloud services



Azure Blob Storage

Azure cloud services



Azure Blob Storage

Azure Virtual Machines

Azure cloud services

Azure Blob Storage   Azure Virtual Machines   Azure SQL Database

Microsoft Fabric

- Integrates various Microsoft solutions for enterprise use
- Covers data movement, data science, real-time analytics, business intelligence
- A key service offering by Microsoft

Azure data services

- **Data Lake Storage** (store data before cleaning)

Azure data services

- **Data Lake Storage** (store data before cleaning)
- **Stream Analytics** (real-time analytics)

Data Lake Storage

Stream Analytics

Azure data services

- **Data Lake Storage** (store data before cleaning)
- **Stream Analytics** (real-time analytics)
- **Machine Learning** (train and deploy machine learning models)

Data Lake Storage

Stream Analytics

Machine Learning

Azure customers

Azure case study

Organization: Ottawa Hospital

Needs: Cost-effective and secure disaster recovery solution (continue vital operations after a disaster)

Solution:

- Microsoft IaaS (secure, scalable environment)
- Azure Storage (medical imaging data)
- Azure Site Recovery (automatically deploy recovery processes)

Azure case study

Improvements:

- New secure, up-to-date, policy compliant disaster recovery site
- Compliant with data privacy regulations
- Saved ~50% on disaster recovery costs

[1] https://customers.microsoft.com/

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Google Cloud

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Google Cloud and the market

Google Cloud

- Google Cloud Anthos
- Run hybrid multi-cloud solutions:
  - manage and deploy across several cloud providers
- Market share: **11%**

Google Cloud services

Google Cloud Storage

Google Cloud services

Google Cloud Storage

Google Cloud Compute Engine

Google Cloud services

Google Cloud Storage

Google Cloud Compute Engine

Google Cloud SQL

Google Cloud data services

- **Big Query** (data warehouse)

Google Cloud BigQuery

Google Cloud data services

- **Big Query** (data warehouse)
- **Dataflow** (batch and stream data processing)

Google Cloud BigQuery

Google Cloud Dataflow

Google Cloud data services

- **Big Query** (data warehouse)
- **Dataflow** (batch and stream data processing)
- **AutoML** (machine learning model training and development)

Google Cloud BigQuery

Google Cloud Dataflow

Google Cloud AutoML

Google Cloud customers

HSBC

vodafone

PayPal

Google Cloud case study

Organization: Lush

Needs: Improve e-commerce platform availability and stability during peak loads

Solution:

- Migrate entire global infra to Google Cloud
- Google Cloud Compute Engine (quickly test and provision environments during migration)
- Customer and product data on Google Cloud SQL

Google Cloud case study

Improvements:

- No outage during Boxing Day
- 40% reduction in hosting costs
- Later deployed an image recognition app to provide information on their product and reduce plastic packaging on Google Cloud AI platform

[1] https://cloud.google.com/customers/lush/

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Congratulations!

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Chapter 1 - Introduction to cloud computing

- How cloud computing **works**
- Why it is **powerful**
- Main **service models** (IaaS, PaaS, SaaS)

Chapter 2 - Cloud strategies

- **Deployment models** (private, public, and hybrid)
- **Regulations**
- Cloud **roles**

Chapter 3 - The cloud infrastructure market

- Market's **major players**
- Their **offerings**
- Their **customers**

Next steps

- **Explore** further to find the right service for you
  - Talk with cloud expert
  - Explore the cloud provider websites
  - Get in touch with cloud providers

DataCamp's cloud-focused **courses**

- Introduction to Azure
- Introduction to AWS
- Introduction to GCP

Thank you!

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